

MIT Open Learning Library — Standard Course Extraction

MITx 8.370.1x — Complete evidence package: SU1 + SU2 + SU3

Evidence boundary: MIT course evidence supplied through course structure, screenshots, accepted answers, hints and discussion is kept separate from QAI interpretation. Primitive candidates are not production claims. Exact complete page/question counts and notebooks are not asserted where they were not supplied.

1. Course Identity

Field	Captured information
Course / ID	MITx 8.370.1x
Course scope	Foundational quantum-computation course evidence covering SU1, SU2 and SU3 processed in this session.
Term / version	1T2018 (from supplied MIT OLL URLs)
MIT OLL URL	https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/
Primary domain	Introduction to quantum computation; classical Boolean logic and reversibility; qubits; quantum mechanics; measurement; tensor products; quantum correlations and quantum weirdness.
Course purpose	Build a foundation for quantum computation from classical Boolean/reversible computation through quantum states, unitary evolution, measurement, tensor products and quantum measurement phenomena.
Source material available	Course introduction, SU1/SU2/SU3 hierarchy information, concept-question screenshots, PS#1/PS#2/PS#3 screenshots, accepted answers, mathematical hints and solution discussions.
Extraction status	Complete for the supplied SU1–SU3 evidence package. Full independent page-by-page crawl, exact total question counts and executable notebooks remain pending/not supplied.

2. Course Fidelity and Source Structure

The MIT terminology is preserved at the evidence layer; QAI interpretation appears only in the extraction/mapping sections.

- SU1: Quantum and classical computing fundamentals — Lectures U1.1/U1.2; PS#1 - Reversible Circuits.
- SU2: Quantum mechanics: qubits and tensor products — Lectures U1.3/U1.4; PS#2 - Quantum Mechanics.
- SU3: Quantum measurement and quantum weirdness — concept questions and PS#3 - Quantum Measurements.
- Problem sets are retained as first-class evidence; the three explicitly identified SU1 implementation problems are preserved.
- No executable notebook/IPYNB was supplied for SU1, SU2 or SU3 in the evidence package.

3. Course Map

Subunit	Lectures / content	Problem set	Evidence status
SU1	U1.1 History and	PS#1 - Reversible Circuits	Complete for supplied

	development of quantum computation; U1.2 Classical computation and reversibility		evidence
SU2	U1.3 Quantum Mechanics I – Qubits; U1.4 Quantum Mechanics II – measurement and tensor products	PS#2 - Quantum Mechanics	Complete for supplied evidence
SU3	Quantum measurement, entanglement, quantum correlations/weirdness; supplied concept questions	PS#3 - Quantum Measurements	Complete for supplied evidence

4. Module / Subunit Record — SU1

Field	SU1 record
Subunit	SU1: Quantum and classical computing fundamentals
Topics	History/development of quantum computation; classical computation; Boolean functions; decision problems; reversibility; fault tolerance; reversible gates; quantum-state foundations.
Important concepts	Universality; Boolean-function cardinality; decision-problem framing; fault tolerance; reversibility; ancilla; compute/use/uncompute; quantum-state equivalence/global phase; Bloch sphere; unitarity; measurement probability; tensor products; circuit-to-unitary composition.
Problem set	PS#1 - Reversible Circuits
Solved implementation problems	Reversible two-four-three swap; Controlled-controlled swap; Reversible two-input demultiplexer.
Reference	Nielsen and Chuang, Chapter 1 (optional reading supplied in course introduction).

4.1 SU1 — Solved evidence and extracted patterns

- Classical circuit universality: circuit models can simulate Turing machines; decision problems provide a foundation for complexity reasoning.
- Fault-tolerance mechanisms: massive redundancy, checkpointing and error correction were identified as classical mechanisms; small imperfections can cascade in reversible physical implementations.
- Boolean-function cardinality: n -input Boolean circuits/functions have 2^{2^n} possible Boolean functions.
- Reversible computation: reversible gates preserve information and require careful handling of ancilla/garbage; reversible constructions expose compute/use/uncompute patterns.
- Reversible gate hierarchy: NOT, CNOT, Toffoli and Fredkin gates were used as primitives; the in-place reversibility question distinguishes what can be constructed without garbage/extra ancilla.
- PS#1 evidence: reversible two-four-three swap; controlled-controlled swap; and two-input demultiplexer were entered and accepted in the MIT OLL interface.
- The SU1 extraction identified composition/decomposition, circuit/operator/matrix translation and structured tensor representation as recurring QAI-relevant patterns.

5. Module / Subunit Record — SU2

Field	SU2 record
Subunit	SU2: Quantum mechanics: qubits and tensor products
Lectures	U1.3 Quantum Mechanics I – Qubits; U1.4 Quantum

	Mechanics II – measurement and tensor products.
Important concepts	Global phase; Bloch sphere; unitary condition; eigenvalues/eigenstates; Pauli matrices; basis transformations; phase and Hadamard gates; single-qubit rotations; measurement; tensor products; controlled operators; CNOT/CZ; operator conjugation.
Problem set	PS#2 - Quantum Mechanics
Solved PS evidence	Pauli matrices/eigenvalues/eigenstates; basis changes; quantum gates; single-qubit rotations; Bloch/axis-angle representation; tensor products of states; tensor products of matrices/CNOT and conjugation.
Notebooks	None supplied.

5.1 SU2 — Previously captured concept questions

- Equivalent quantum states: states differing only by a global phase represent the same physical quantum state.
- Bloch-sphere states: $|0\rangle$, $|1\rangle$, and $\pm x/\pm y$ axis states were mapped to geometric positions.
- Valid unitary matrices: $U^\dagger U = I$ is the validity condition.
- Hadamard rotation axis: the axis is proportional to $(1/\sqrt{2}, 0, 1/\sqrt{2})$.
- Measurement outcome probabilities follow amplitude magnitudes via the Born rule.
- Tensor product of X and Y produces a structured 4×4 operator.
- A quantum circuit can be represented by multiplication of its gate matrices in execution order.

5.2 SU2 — PS#2 extracted models

Evidence item	Mathematical / procedural model	QAI-relevant pattern
Pauli eigenstructure	$A \psi\rangle = \lambda \psi\rangle$	Eigenstructure / natural measurement basis
Basis change	$A_{\text{new}} = S^\dagger A_{\text{old}} S$	Representation transformation without semantic change
Gate composition	$U_{\text{total}} = U_m \dots U_2 U_1$	Sequential operator composition
Operator conjugation	$A' = U A U^\dagger$	Transform an operator under a unitary
Axis-angle	$U = \exp(-i \theta \mathbf{n} \cdot \boldsymbol{\sigma} / 2)$	Geometric operator representation
State tensor product	$ a\rangle \otimes b\rangle$	Composite-state construction
Matrix tensor product	$A \otimes B$ as block matrix $[A_{ij} B]$	Structured composite operator
Controlled operator	$ 0\rangle\langle 0 \otimes I + 1\rangle\langle 1 \otimes X$	Projector-selected conditional operation
CNOT/CZ relation	Conjugation by $I \otimes H$ maps CNOT to controlled-Z	Gate-family transformation / equivalence

6. Module / Subunit Record — SU3

Field	SU3 record
Subunit	SU3: Quantum measurement and quantum weirdness
Evidence	Concept questions and PS#3 - Quantum Measurements screenshots supplied and accepted/correct.
Core themes	Hermitian/unitary operators; two-qubit spin observables; Bell states; expectation values; CHSH correlations; commutation and simultaneous diagonalization; measurement; indirect measurements; entanglement; GHZ game; general-basis measurement.
Problem set	PS#3 - Quantum Measurements
Notebooks	None supplied.

6.1 SU3 — Concept-question evidence

Question / topic	Captured result / insight
Unitary and Hermitian matrices	Only U unitary $\Rightarrow U^\dagger U = I$ Hermitian is universally true among

	the four statements; Hermitian and unitary are distinct properties.
Two-qubit spin operators	$X \otimes I + I \otimes X$, $X \otimes X$, and $Y \otimes I + I \otimes Y$ map to the supplied Hermitian 4×4 matrices.
Bell-state expectation values	Expectation values $\langle \psi S \psi \rangle$ were evaluated for Bell states and total-spin observables; supplied examples gave zero for the displayed cases.
CHSH game	All four Bell states can support a successful quantum protocol; quantum success $\cos^2(\pi/8) \approx 0.854$ exceeds the classical maximum 0.75.
Commuting matrices / simultaneous diagonalization	Commuting Hermitian matrices are simultaneously diagonalizable by a unitary; simultaneously diagonalizable Hermitian matrices commute.

6.2 SU3 — PS#3 solved evidence

PS#3 item	Captured solution / result	Pattern
Q1 Measurements	After $U = e^{i(\theta/2)Y}$, $P(0 0) = \cos^2(\theta/2)$; $P(0 +) = (1 + \sin \theta)/2$; equal classical mixture gives 1/2.	Measurement-basis rotation; coherent superposition vs classical mixture
Q2 Bell states	$CNOT(H \otimes I) 10\rangle$ produces $ \Phi\rangle = (00\rangle - 11\rangle)/\sqrt{2}$; subsequent local transformation produces $ \Psi\rangle = (01\rangle - 10\rangle)/\sqrt{2}$.	Entangled-state construction from gates with circuit provenance
Q3 Entanglement definition	Product states are unentangled; non-factorizable states are entangled. A state with several ket terms can still be separable.	Separable/entangled state analysis
Q4 Indirect measurements	For $U = X$ and $ \psi\rangle = a 0\rangle + b 1\rangle$, $P(0) = (a+b)^2/2$ and conditional final state is $(0\rangle + 1\rangle)/\sqrt{2}$. General eigenphase form gives $[1 + a^2 \cos \theta_1 + (1 - a^2) \cos \theta_2]/2$.	Ancilla-assisted indirect measurement and phase/operator information
Q5 GHZ classical strategy	Deterministic classical strategies cannot satisfy all GHZ constraints; maximum classical winning probability is 0.75.	Classical correlation baseline
Q6 GHZ quantum strategy	Shared 3-qubit entangled state plus input-dependent local H operations yields winning probability 1 for the game.	Multi-party entanglement-assisted correlation
Q7 General-basis measurement	For $n \cdot \sigma$ measured on $ 0\rangle$, $P(+1) = (1 + n_z)/2$.	Axis/basis-independent measurement interface

7. Course-Level Algorithms and Mathematical Models

Model / procedure	Core expression / structure	Evidence
Reversible circuit composition	Sequence of reversible gates with no information loss	SU1 PS#1
Quantum state parameterization	$ \psi\rangle = \cos(\theta/2) 0\rangle + e^{i\phi} \sin(\theta/2) 1\rangle$	SU2
Unitary validation	$U^\dagger U = I$	SU2 / SU3
Eigenstructure	$A \psi\rangle = \lambda \psi\rangle$	SU2 / SU3
Basis transformation	$A' = S^\dagger A S$	SU2
Operator conjugation	$A' = U A U^\dagger$	SU2
Axis-angle unitary	$U = \exp(-i\theta \cdot n \cdot \sigma/2)$	SU2
Born-rule measurement	$P(i) = \alpha_i ^2$	SU2 / SU3
Expectation value	$\langle \psi O \psi \rangle$	SU3 Bell/spin evidence
Tensor composition	$A \otimes B$; $ a\rangle \otimes b\rangle$	SU1 / SU2 / SU3

Controlled operator	Projector-selected tensor terms	SU2
General measurement	$n \cdot \sigma$ with eigenvalues ± 1	SU3
CHSH correlation benchmark	Quantum ≈ 0.854 vs classical ≤ 0.75	SU3
GHZ correlation benchmark	Quantum 1 vs classical ≤ 0.75	SU3

8. Recurring Patterns Across SU1–SU3

- Composition/decomposition: small primitive operators and gates compose into larger reversible, unitary and tensor-structured systems.
- Representation transformation: the same operation can be represented as a circuit, matrix, operator, eigenbasis or Bloch-sphere geometry.
- Tensor/block structure: subsystem operators expand into structured composite operators; block structure can be retained instead of treating the expanded matrix as opaque.
- State/operator separation: quantum state, transformation operator, observable and measurement outcome are distinct computational objects.
- Eigenstructure as an interface: eigenvalues/eigenstates provide natural representations for observables, measurement and phase information.
- Conditional execution: controlled gates and projector-selected operators express conditional behavior without abandoning reversibility/unitarity.
- Measurement as an information-extraction layer: probabilities, expectation values, indirect measurements and correlation tests are downstream of state/operator construction.
- Classical-vs-quantum benchmark pattern: establish a classical bound, implement the quantum protocol, then compare measured success/correlation.
- Coherence matters: a coherent superposition can respond differently to a unitary/measurement sequence than a classical statistical mixture.
- Entanglement is structural: it is determined by factorization properties, not simply by the number of basis terms.

9. QAI Primitive Candidates

These are candidates derived from the supplied evidence. They are not production capabilities and require independent implementation and validation.

Candidate	Evidence	Potential QAI role
Reversible Gate Composition	SU1 PS#1	Reversible circuit construction and round-trip validation
State / Operator Representation	SU1–SU3	Common representation layer for states, operators and observables
Eigenstructure Analyzer	SU2/SU3	Eigenvalues, eigenvectors, measurement bases and operator analysis
Basis Transform	SU2	Representation conversion
Unitary Composition / Validation	SU2/SU3	Gate composition and $U^\dagger U$ validation
Operator Conjugation	SU2	Transform observables/operators under unitary evolution
Bloch / Axis-Angle Conversion	SU2	Geometric state/operator representation
Tensor Composition	SU1/SU2/SU3	Composite states/operators and block structures
Controlled Operator Builder	SU1/SU2	Projector-based conditional operators
Structured Tensor / Block Analyzer	SU1/SU2	Retain and analyze structured composite operators; QAI-derived hypothesis
Expectation Value	SU3	Observable evaluation

General Measurement	SU3	Axis/basis-independent measurement interface
Indirect Measurement / Phase Probe	SU3; related earlier AQSE material	Ancilla-assisted operator/phase information extraction
Entanglement Analyzer	SU3	Separability / entanglement classification
Quantum Correlation Benchmark	SU3 CHSH/GHZ	Compare quantum correlation protocol against classical bound

10. QAI Architecture Mapping

QAI layer	Course-derived contribution
QAI Language / semantic layer	Operations can be described independently of a particular matrix expansion; candidates include state, observable, measurement, tensor and controlled-operation semantics.
Representation / IR	Strong evidence for circuit $\leftrightarrow$ operator $\leftrightarrow$ matrix $\leftrightarrow$ basis $\leftrightarrow$ geometric representation transformations.
QAI Runtime	Candidate execution of reversible/quantum circuits, unitary transformations, tensor composition and measurement operations.
Hybrid Runtime	Classical analysis can coordinate representation conversion, validation, benchmark comparison and measurement post-processing; not directly implemented by the course.
QAI Control Plane	Potential policies for circuit construction, resource/ancilla handling and execution strategy; future mapping.
Quantum Control Plane	Potential management of basis selection, state preparation, controlled operations and measurements; future mapping.
Resource Registry	Potential tracking of qubit dimensions, tensor factors, operator sizes, ancilla and execution requirements.
Capability Registry	Candidate capabilities: eigenstructure, unitary composition, basis transformation, tensor composition, controlled operators, measurement and correlation benchmarks.
QAI Labs	Natural validation location for reversible circuits, state/operator models, tensor composition, measurement and CHSH/GHZ benchmarks.
QAI Product Foundry	Potential future source of validated reusable quantum/operator components; not a direct course deliverable.

11. Validation and Promotion Plan

- Reference implementation: implement the identified mathematical primitives independently of the MIT answer-entry interface.
- Simulator validation: normalization, unitarity, eigenpair equations, basis round trips, Bloch rotations, tensor dimensions/order, controlled-gate truth tables, measurement probabilities and expectation values.
- Independent problem validation: add randomized matrices/states, alternative bases, additional controlled gates, multi-qubit tensor cases and independent measurement/correlation tests.
- Benchmarking: compare dense-matrix and structured/tensor representations for correctness, memory and execution scaling where applicable.
- Correlation validation: reproduce CHSH classical bound/quantum strategy and GHZ classical bound/quantum strategy with independent implementations.

- Hardware/QPU validation: optional later stage for physical gate and measurement behavior; not required to establish the mathematical evidence.
- Promotion gate: Candidate → Specification → Reference Implementation → Validation → Reusable Primitive → Enterprise Library.
- No production primitive should be promoted from this educational course alone.

12. Evidence Classification

Classification	Course-level result
Educational knowledge	Classical/reversible computation; qubits; Bloch sphere; unitary and Hermitian operators; measurement; tensor products; Bell/GHZ/CHSH concepts.
Mathematical model	Boolean functions; unitary/Hermitian conditions; eigenstructure; basis transforms; tensor products; Born rule; expectation values; general-basis measurement.
Algorithm / procedure	Reversible circuit constructions; gate composition; basis transformation; operator conjugation; tensor construction; indirect measurement; correlation-game protocols.
Pattern	Composition/decomposition; structured representation; conditional operations; measurement abstraction; classical-vs-quantum benchmark.
Implementation evidence	PS#1, PS#2 and PS#3 displayed accepted/correct answers in the supplied MIT OLL screenshots.
Reusable capability	Candidate only; independent implementation and validation required.
Future capability	Structured tensor scoping, optimized measurement grouping, runtime integration and broader QAI services remain future mappings/hypotheses.
Research	Not claimed solely from this course.

13. Cross-Course / Prior-Course Convergence

- SU1 → SU2: reversible circuits and circuit/operator/matrix representation are strengthened by formal quantum states, unitaries, eigenstructure, basis changes and tensor products.
- SU2 → SU3: state/operator/tensor foundations become measurement, expectation, entanglement and correlation mechanisms.
- SU1 → SU3: reversible/controlled composition provides the construction substrate for measurement and multi-qubit correlation protocols.
- Relation to prior MIT 8.371.3x material: controlled-U, eigenphase probing, measurement, estimation and feedback are conceptually compatible with the measurement/operator foundations here; full cross-course validation remains pending.
- The strongest emerging architectural chain is: state → operator → composition → transformation → measurement → estimation/correlation → benchmark/feedback.

14. Course-Level Retention Summary

MITx 8.370.1x SU1–SU3 should be retained as a foundational evidence package for the QAI stack. Its strongest value is the progressive construction of a mathematical and operational substrate: classical/reversible primitives; quantum state and operator representations; eigenstructure and basis transformation; tensor/block composition; controlled operators; measurement; expectation values; entanglement; indirect measurement; and quantum correlation benchmarks.

The course provides especially strong evidence for a layered QAI representation and operator-algebra model in which semantic operations, circuit descriptions, matrices, tensor factors, observables, measurements and validation results remain related but distinct representations.

The QAI Enterprise Library contribution at this stage is a set of evidence-backed candidate abstractions, not production-ready primitives. Cross-course recurrence, independent reference implementations, simulator validation and later hardware validation should be used before promotion.

15. Extraction Status

Area	Status	Notes
Course hierarchy	Complete for supplied evidence	SU1, SU2 and SU3 hierarchy captured; full independent crawl not performed.
Modules / subunits	Complete	SU1–SU3 captured.
Lectures / content	Captured	U1.1–U1.4 identified; SU3 lecture-page inventory not independently supplied.
Problem sets	Complete for supplied evidence	PS#1, PS#2 and PS#3 evidence captured.
Solved questions	Complete for supplied evidence	Captured concept questions plus supplied PS items.
Notebooks	Pending / not supplied	No executable IPYNB supplied in this evidence package.
Algorithms / procedures	Captured	Reversible construction, operator algebra, tensor construction, measurement and correlation procedures.
Mathematical models	Captured	States, unitaries, eigenstructure, tensor products, Born rule, expectation values, Bell/GHZ correlations.
Patterns	Captured	Recurring patterns identified and separated from MIT source evidence.
Primitive candidates	Candidate	No production promotion.
QAI mapping	Captured	Architecture relationships recorded.
Cross-course mapping	Partially captured	SU1–SU3 convergence established internally; broader validation with later courses remains pending.
Validation	Planned	Reference implementation, simulator, independent and later hardware validation required.
Final course review	Complete for supplied evidence	Course-level package consolidated.

16. Evidence and Provenance

- MIT OLL course URL: <https://openlearninglibrary.mit.edu/courses/course-v1:MITx+8.370.1x+1T2018/course/>
- SU1 source evidence: supplied course structure, screenshots and PS#1 accepted answers; optional Nielsen and Chuang Chapter 1 reference from course introduction.
- SU2 source evidence: supplied SU2 hierarchy, previously solved CQs, PS#2 screenshots and accepted answers.
- SU3 source evidence: supplied concept-question and PS#3 screenshots with accepted answers and solution discussion in this session.
- Evidence limitation: this package is based on supplied source material and captured course interactions; it is not a complete independent crawl of every MIT OLL page.
- Notebook limitation: no executable notebooks were supplied for this course evidence package.

17. Master Working Rule

PRESERVE → EXTRACT → ANALYZE → GENERALIZE → VALIDATE → MAP → PROMOTE

The course is retained both as a solved-course knowledge set and as a source of candidate reusable QAI patterns. Promotion to the Enterprise Library remains gated by independent validation.

Appendix A — Core QAI Primitive Family Emerging from 8.370.1x

- State representation / canonicalization
- Unitary and Hermitian validation
- Eigenstructure analysis
- Basis transformation
- Unitary/gate composition
- Operator conjugation
- Bloch / axis-angle conversion
- Tensor state/operator composition
- Controlled operator construction
- Structured tensor/block analysis
- Measurement probability evaluation
- Expectation-value evaluation
- General-basis measurement
- Indirect measurement / phase probing
- Entanglement / separability analysis
- Quantum correlation benchmark

End of complete 8.370.1x SU1–SU3 extraction — supplied evidence package.